

Parle Tilak Vidyalaya Association's
MULUND COLLEGE OF COMMERCE (AUTONOMOUS)
T.Y. B.Sc. Information Technology
Semester V(Autonomous)

Course: Artificial Intelligence
Duration: 40 min.

Date: 12th December, 2024
Max Marks: 20

Answer any four of the following:

20

1. What is Artificial Intelligence? Explain its four applications in detail.
2. Formulate 8-puzzle problem for the following scenario and also mention the steps and path cost required to reach goal state.

1	2	3
4	8	-----
7	6	5

Initial State

1	2	3
4	5	6
7	8	-----

Goal State

3. Describe the Minimax algorithm with an example.
4. Convert the below given Facts into FOL and Prove that "Colonel is a Criminal" using Backward-Chaining.
 - i) It is crime for an American to sell weapons to the enemy of America.
 - ii) Country Nono is an enemy of America.
 - iii) Nono has some Missiles.
 - iv) All the Missiles were sold to Nono by Colonel.
 - v) Missile is a weapon.
 - vi) Colonel is American.
5. Explain multi-agent planning with different types of strategies.
6. Differentiate between Propositional logic and Predicate Logic.